



Bias: From questions to conclusions

Let's kick things off by traveling back in time, well, in our minds at least. My real time machine's in the shop. Imagine you're back in middle school and you've entered a project for the science fair. You worked hard for weeks perfecting every element and they're about to announce the winners. You close your eyes, take a deep breath, and you hear them call your name for second place. Bummer, you really wanted that first-place trophy, but hey, you'll take the ribbon for recognition. The next day you learn the judge was the winner's uncle. How is that fair!? Can he really be expected to choose a winner fairly when his own family member is one of the contestants? He's probably biased! Maybe his niece deserved to win and maybe not. But the point is: it's very easy to make a case for bias in that scenario. This is a super-simple example, but the truth is, we run into bias all the time in everyday life. Our brains are biologically designed to streamline thinking and make quick judgments. Bias has evolved to become a preference in favor of or against a person, group of people, or thing. It can be conscious or subconscious. The good news is once we know and accept that we have bias, we can start to recognize our own patterns of thinking and learn how to manage it. It's important to know that bias can also find its way into the world of data. Data bias is a type of error that systematically skews results in a certain direction. Maybe the

questions on a survey had a particular **slant** to influence answers, or maybe the sample group wasn't truly representative of the population being studied. For example, if you're going to take the median age of the US patient population with health insurance, you wouldn't just use a sample of Medicare patients who are 65 and older. **Bias can also happen if a sample group lacks inclusivity.** For example, people with disabilities tend to be under-identified, under-represented, or excluded in mainstream health research. **The way you collect data can also bias a data set.** For example, if you give people only a short time to answer questions, their responses will be rushed. When we're rushed, we make more mistakes, which can affect the quality of our data and create biased outcomes. **As a data analyst, you have to think about bias and fairness from the moment you start collecting data to the time you present your conclusions.** After all, those conclusions can have serious implications. Think about this: it's been acknowledged that clinical studies of heart health tend to include a lot more men than women. This has led to women failing to recognize symptoms and ultimately having their heart conditions go undetected and untreated. That's just one way bias can have a very real impact. While we've come a long way in recognizing bias, it still led to you losing out to the judge's niece at that science competition. It's still influencing business decisions, health care choices and access, governmental action, and more. So we've still got work to do. Coming up, we'll show you how to identify bias in the data itself, and explore some scenarios when you may actually benefit from it.